Ejercicios básicos de R – S3

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Lunes 27 de abril de 2026

Cargar los virus

```
virus <- c(
  "FJ797658.1",
  "M10098.1",
  "M59389.1",
  "NR_024570.1",
  "MW558256.1",
  "NC_012059.1"
)
```

1. Cargar las librerías que necesitaremos

```
library(Biostrings)
library(seqinr)
library(adegenet)
library(ape)
library(ggtree)
library(DECIPHER)
library(viridis)
library(ggplot2)
```

2. Obtener las secuencias

```
covid_seq <- ape::read.GenBank(
  virus,
  species.names = TRUE,
  as.character = FALSE
)
```

3. Estructura del DNABin

```
## 6 DNA sequences in binary format stored in a list.
##
```

```
## Mean sequence length: 3710.167
##    Shortest sequence: 300
##     Longest sequence: 16388
##
## Labels:
## FJ797658.1
## M10098.1
## M59389.1
## NR_024570.1
## MW558256.1
## NC_012059.1
##
## Base composition:
##     a     c     g     t
## 0.306 0.261 0.168 0.264
## (Total: 22.26 kb)
```

4. Concentrar en un archivo las secuencias

```
ape::write.FASTA(
  covid_seq,
  file = "covid_seq.fasta"
)
```

5. Cargar las secuencias

```
dna <- Biostrings::readDNAStringSet("covid_seq.fasta")
```

6. Alineamiento de secuencias

```
alignment <- DECIPHER::AlignSeqs(dna)
```

```
## Determining distance matrix based on shared 11-mers:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Clustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Aligning Sequences:
## ================================================================================
##
## Time difference of 3.5 secs
##
```

```
## Iteration 1 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 2.97 secs
##
## Iteration 2 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 0 secs
##
## Refining the alignment:
## ================================================================================
##
## Time difference of 9.61 secs
```

7. Visualizar el resultado

```
alignment
```

```
## DNAStringSet object of length 6:
##     width seq                                               names
## [1] 17654 ----------...---------- FJ797658.1
## [2] 17654 CCGTCCGTCCGTCGTCCTCCTCG...---------- M10098.1
## [3] 17654 ----------...---------- M59389.1
## [4] 17654 ----------...---------- NR_024570.1
## [5] 17654 ----------...---------- MW558256.1
## [6] 17654 ----------...ACAAATAATTTCCTACCCCATAA NC_012059.1
```

8. Guardar el resultado

```
Biostrings::writeXStringSet(
  alignment,
  filepath = "covid_alignment.fasta"
)
```

9. Obtener el archivo

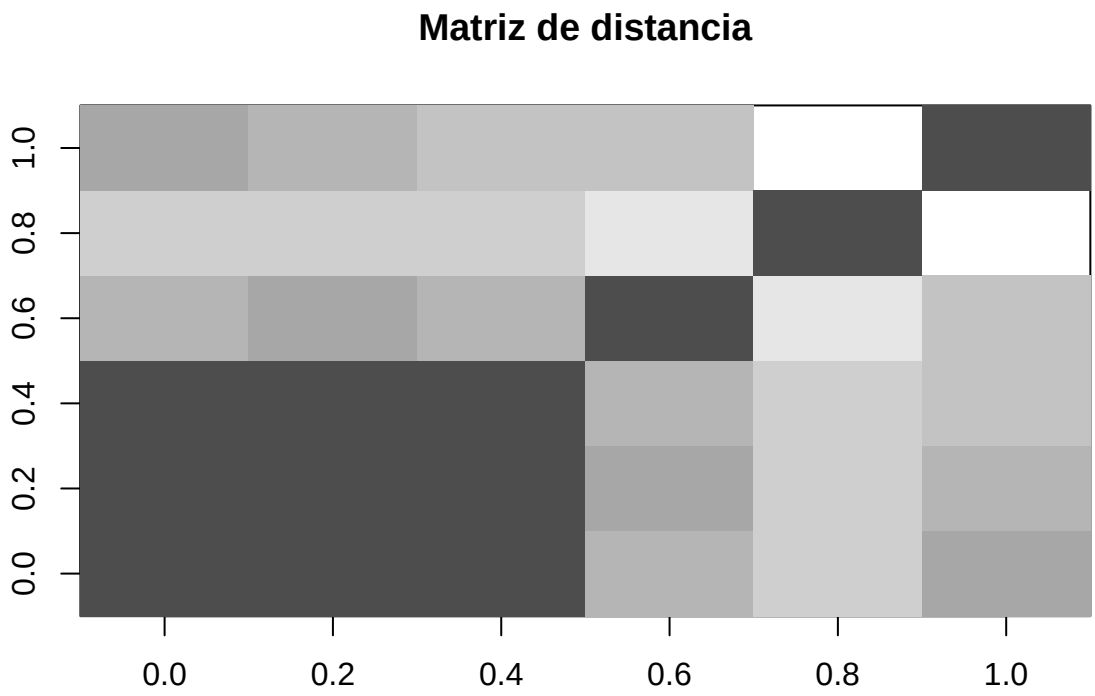
```
aligned <- Biostrings::readDNAStringSet("covid_alignment.fasta")
```

10. Matriz de distancia

```
aligned_dnabin <- ape::as.DNAbin(aligned)
dist_matrix <- as.matrix(ape::dist.dna(
  aligned_dnabin,
  pairwise.deletion = TRUE
))
```

11. Visualizar Matriz

```
image(
  as.matrix(dist_matrix),
  main = "Matriz de distancia",
  col = gray.colors(10)
)
```

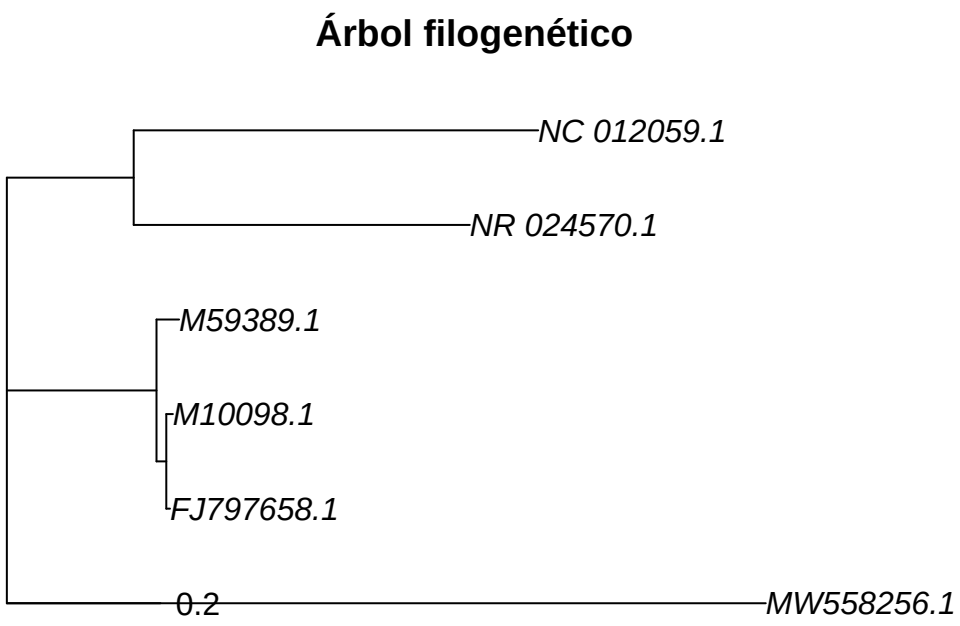


12. Crear árbol

```
tree <- ape::njs(dist_matrix)
```

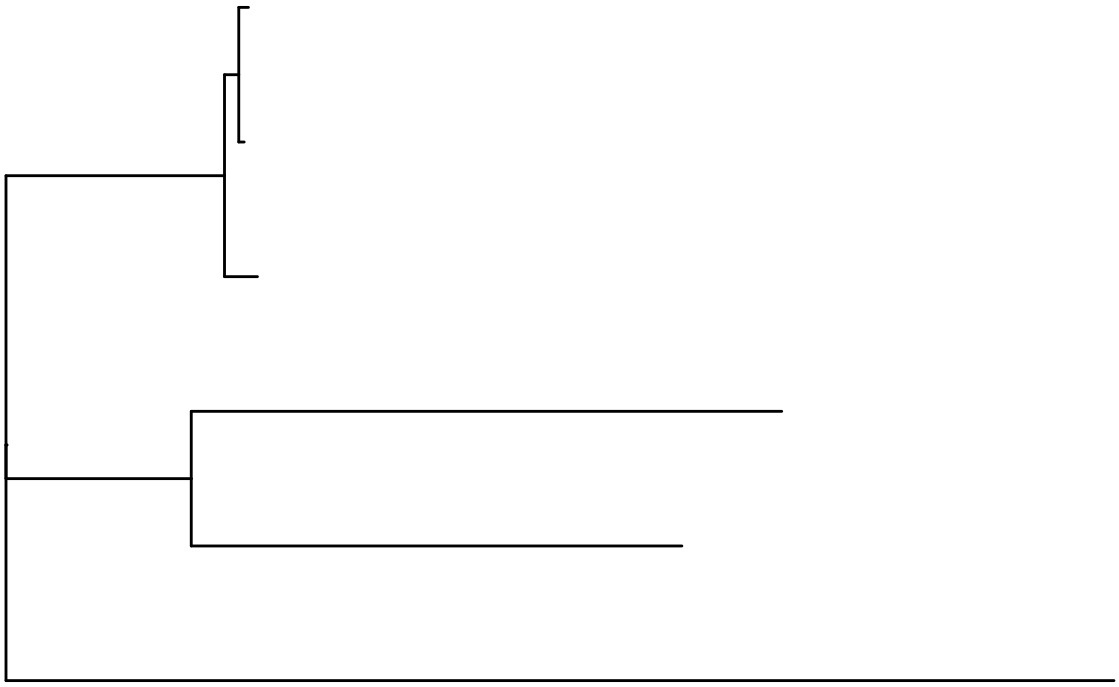
13. Plot del árbol

```
plot(
  tree,
  main = "Árbol filogenético"
)
ape::add.scale.bar()
```



14. Plot con ggtree

```
options(ignore.negative.edge = TRUE)
ggtree::ggtree(tree)
```



15. Visualizar alineamiento

```
aligned_table <- data.frame(
  Secuencia = names(alignment),
  Alineamiento = as.character(alignment)
)

aligned_table
```

```
##                Secuencia
## FJ797658.1   FJ797658.1
## M10098.1       M10098.1
## M59389.1       M59389.1
## NR_024570.1 NR_024570.1
## MW558256.1   MW558256.1
## NC_012059.1 NC_012059.1
##
## FJ797658.1  ----------------------------------------------------------------------------------------------------
## M10098.1    CCGTCCGTCCGTCGTCCTCCTCGCTTGCGGGGCGCCGGGCCCGTCCTCGAGCCCCCNNNNNCCGTCCGGCCGCGTCGGGGCCTCGCCG
## M59389.1    ----------------------------------------------------------------------------------------------------
## NR_024570.1 ----------------------------------------------------------------------------------------------------
## MW558256.1  ----------------------------------------------------------------------------------------------------
## NC_012059.1 ----------------------------------------------------------------------------------------------------
```

16. Combinar árbol y alineamiento

```
ggtree::ggtree(tree) +
  ggtree::geom_tiplab()
```

